Amit Singh

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EDUCATION

National Institute of Technology, Jamshedpur **2023 – 2027**
Bachelor of Technology *CGPA: 7.56/10*

EXPERIENCE

Research Intern - Slurry Wear Rate Prediction **July – Oct 2026**
Python, Pandas, Scikit-learn, Plotly

- Built an end-to-end ML pipeline for **slurry erosion wear-rate prediction**: data preprocessing, interaction features, categorical encoding, and EDA.
- Benchmarked **10 regression configurations** with grouped cross-validation; tuned **Extra Trees, Random Forest, and Gradient Boosting**.
- Reached a preliminary test $R^2 = 0.994$ and **RMSE = 0.250** (Extra Trees); used permutation importance and residual analysis to identify key predictors and error sources.

PROJECTS

Multi-Agent LLM Orchestration Platform | LangGraph, FastAPI, Groq, Gemini, AWS, Streamlit Link | May 2026

- Designed a **LangGraph multi-agent framework** where a **zero-shot Router Agent** assigns confidence scores to select expert agents and an **Aggregator Agent** synthesizes their responses.
- Built a **confidence-aware scatter-gather pipeline** (`asyncio.gather`) across **3 concurrent LLM agents**, with routing latency under **800 ms** and end-to-end orchestration under **4 s**.
- Integrated **Groq LLaMA, Gemini, Tavily, Wikipedia, and ArXiv** as tools with confidence-threshold filtering; deployed on **AWS ECS Fargate** via Docker and Jenkins.

SportShield | FastAPI, Next.js, Google Cloud Platform, Firebase, Vertex AI Link | April 2026

- Built a **full-stack platform** (**FastAPI**, **Next.js**, **9 Google Cloud services** incl. Cloud Run, Firestore, Vertex AI, Cloud Vision) that detects unauthorized use of creators' photos, videos, and audio online.
- Built a detection engine that fingerprints uploads and finds copies via reverse image search and AI-based matching, ranked by confidence; automated **DMCA takedowns**, watermarking, **C2PA** signing, and a Chrome extension.

Medical RAG Chatbot | Flask, LangChain, HuggingFace, Docker, AWS Link | Sep 2025

- Designed an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline over **2,800+ pages** of medical PDFs; implemented document parsing, chunking, and **semantic retrieval** using FAISS for contextual question answering.
- Generated **vector embeddings** with **HuggingFace Transformers**; optimized retrieval to achieve production-grade benchmarks of **95% Recall@5** and **120 ms avg latency**.
- Deployed **Flask** backend via Docker on **AWS (ECR/EC2)** with full **CI/CD** pipeline for scalable, low-latency inference.

TECHNICAL SKILLS

- **ML/LLM:** PyTorch, TensorFlow, Scikit-learn, RAG, LangChain, LangGraph, MCP, Vector Embeddings, Agentic AI
- **Programming:** Python, C++, SQL (MySQL), Pandas; Data Structures & Algorithms, OOP, DBMS
- **Backend & Cloud:** FastAPI, Flask, REST APIs, Docker, AWS (ECS, ECR, EC2), Google Cloud Platform
- **DevOps & MLOps:** Git, GitHub Actions, Jenkins, CI/CD, MLflow, DagsHub

ACHIEVEMENTS & CERTIFICATIONS

- **2nd Runner-Up, ZS Campus Beats**: placed among **7,000+ participants** across India in business case-solving and analytical reasoning.
- Selected for the **Grand Finale of Meta PyTorch × Scaler OpenEnv Hackathon** among top teams out of **70,000+ registrations**. Link
- **Top 106 globally, Google Solution Challenge**: out of **85,000+ developers** and **6,700+ submissions**. Link
- Achieved **Top 25 Percentile** in the IIT Guwahati C&A Club competition among 500+ participants. Certificate

POSITIONS OF RESPONSIBILITY

- **Placement Coordinator**, Training and Placement Cell, NIT Jamshedpur *May 2025 – Present*
- **Core Team Member**, Team Phoenix (Aeromodelling Club, NIT Jamshedpur) *Jan 2024 – Present*
  Represented NIT Jamshedpur at **IIT Bombay Techfest** (**2nd Runner-Up**), **IIT Kharagpur Kshitij**, and **SAE Aero Design West 2025**, California. Certificate